

MSE 360: Kinetic Processes in Materials
S2025
Midterm Exam 1

Points: ____/100

Name (print): _____

I have neither given nor received unauthorized aid on this test.

_____ (signature)

No notes, books, cell phones, or information stored in calculator memories may be used. Suspicion of cheating will result in the filing of a Report of Academic Integrity Violation. All of your work must be written on these pages and turned in.

Useful Constants

$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$	$1 \text{ eV/atom} = 23.06 \text{ kcal/mol}$
$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$	$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$
$R = 8.3142 \text{ J mol}^{-1} \text{ K}^{-1}$	$k_B = 8.616 \times 10^{-5} \text{ eV K}^{-1}$
$R = 82.05 \text{ cm}^3 \text{ atm mol}^{-1} \text{ K}^{-1}$	$N_A = 6.022 \times 10^{23} \text{ molecules mol}^{-1}$
$e = 1.602 \times 10^{-19} \text{ C}$	$101,325 \text{ Pa} = 1 \text{ atm}$
